

NATIONAL VICTOR PUBLIC SCHOOL

COMPUTER SCIENCE PROJECT

LIBRARY MANAGEMENT

- NAME : HIMANI DHAWAN
- CLASS : XII-A
- SUBJECT : COMPUTER SCIENCE
- ROLL NO. :
ADMISSION NO. : 10846

ACKNOWLEDGEMENT

- IT IS MY PRIVILEGE TO EXPRESS MY MOST SINCERE REGARDS TO ALL WHO DIRECTLY OR INDIRECTLY HELPED ME TO COMPLETE MY PROJECT WORK. I ACKNOWLEDGE MY HEARTIEST GRATITUDE TO MY COMPUTER SCIENCE TEACHER MS SHEETAL VERMA (PGT IN COMPUTER SCIENCE) FOR HER VALUABLE INPUTS, ENCOURAGEMENT, WHOLE-HEARTED COOPERATION THAT LED ME ALL THE WAY TO ACCOMPLISH THIS PROJECT. I TAKE THIS OPPORTUNITY TO THANK ALL MY TEACHERS, PARENTS & FRIENDS WHO HAVE HELPED ME WITH THEIR VALUABLE SUGGESTIONS, ENCOURAGEMENT & GUIDANCE THAT HAVE BEEN VERY HELPFUL IN VARIOUS PHASES OF THE COMPLETION OF THE 'LIBRARY MANAGEMENT PROJECT'

CERTIFICATE

- THIS IS TO CERTIFY THAT HIMANI DHAWAN OF CLASS/SEC XII – A HAS SUCCESSFULLY COMPLETED THE PROJECT ENTITLED LIBRARY MANAGEMENT PROJECT TO MY SATISFACTION AND SUBMITTED THE SAME DURING THE ACADEMIC YEAR 2023-24.. THE PROJECT IS THE RESULT OF HIS/HER EFFORTS & ENDEAVORS.

•
.....

(SIGNATURE OF THE TEACHER)

INDEX

S.NO.	PARTICULARS	PAGE NO.
1.	HARDWARE AND SOFTWARE REQUIREMENT	
2.	REPORT	
3. 4.	SYNOPSIS FUNCTIONS AND MODULES	
5.	MYSQL CODE	
6.	PYTHON CODE	
7.	RESULT	

HARDWARE AND SOFTWARE REQUIREMENT:

❖ SOFTWARE SPECIFICATION:

- OPERATING SYSTEM: WINDOWS 7
- PLATFORM: PYTHON IDLE 3.7
- DATABASE: MYSQL

❖ HARDWARE SPECIFICATION:

- PROCESSOR : INTEL(R) CORE(TM) I3-7130U CPU @ 2.70GHZ 2.71 GHZ
- INSTALLED RAM: 4.00 GB

SYSTEM TYPE :64-BIT OPERATING SYSTEM, X64-BASED PROCESSOR

REPORT

- The library management deals in management of records. The records include books which are required by different students accordingly. Maintaining these records manually becomes difficult thus we are providing such system a computerized backend. In this system each member is provided with a member code and all the records are maintained in a table along with the member's code. Any information can be accessed according to member-code anytime from the table and also at the time of issuing books.
- The Automated System Have Following Benefits Over Manual System :
 - ☐ Data handling: It captures the information from different sources, presents it systematically and organizes its storage for efficient retrieval.
 - ☐ Quality control: Paper work would totally be eliminated in the new system as failure data is directly fed into system.
 - ☐ System reliability: System is very reliable as no skipping, missing of data is possible.
 - ☐ Maintenance: No data mismatching is possible due to various checks incorporated in the system.



- ❑ Accuracy : The data provided by the system will be accurate as all Processing steps are algorithmic and computer based.

- ❑ Centralized Storage: The data is Processed and stored at central location.

- SAILENT FEATURES OF NEW PROJECT :

- ❑ Integrated approach to the package helps in minimizing manual approach.

- ❑ User friendly package.

- ❑ Timing and retrieval efficiency.

- ❑ In-built validation checks to ensure accuracy.

- ❑ Elimination of paper work.

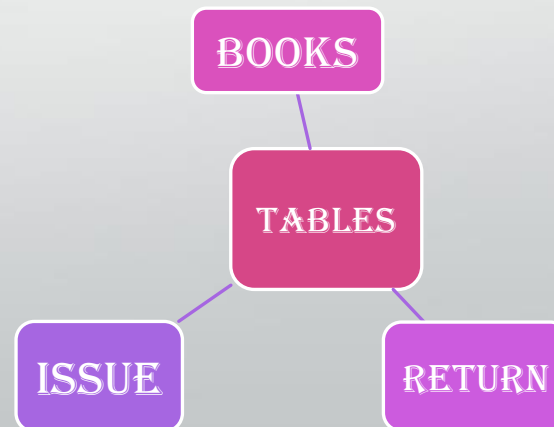
- ❑ Suitable help/error message for better user interface.

- ❑ Updated information.

SYNOPSIS

- A library management system is a web application that is designed to manage all the functions of a library. It helps librarian to maintain the database of new books, and the books that are borrowed by members along with their due date . It helps in smooth functioning of library also it is implementation of a database management system.
- The purpose of library management system is to automate the existing manual system by the help of computerized equipments and full-fledged computer software, Fulfilling their requirements , so that their valuable data can be stored with easy accessing and manipulation of the same. The required software and hardware are easily available and easy to work with.
- FRONT END- PYTHON
BACK END- MYSQL

- Project IS created using python and mysql.
- Database named library is used to make tables for storing information.
- **Tables are:-**
- **Issue** : consists of book name , book code, student name, issue date. Which stores data about book name ,its code , name of student which is issuing books , and date in form of rows and column.
- **Books** : consists of bookname , authorname ,bookcode ,total which stores data about book name ,its author ,its code and its quantity in form of rows columns.
- **Return** : consists of book name , book code, student name, return date. Which stores data about book name ,its code , name of student which is returning books , and date in form of column.



FUNCTIONS AND MODULES

- **MODULES:**

1. **IMPORT MYSQL.CONNECTOR**

- **BY IMPORTING THIS, WE WILL BE ABLE TO ESTABLISH CONNECTION BETWEEN SQL AND PYTHON.**

- **FUNCTIONS:**

1. **CONNECT():**

THIS FUNCTION ESTABLISHES CONNECTION BETWEEN PYTHON AND MYSQL .

2. **CURSOR():**

IT IS A SPECIAL CONTROL STRUCTURE THAT FACILITATE ROW BY ROW PROCESSING OF RECORDS IN THE RESULT SET.

3. **EXECUTE ():**

THIS IS THE FUNCTION USED TO EXECUTE SQL QUERY AND RETRIEVE RECORDS USING PYTHON.

- 4. DEF():
- A FUNCTION IS BLOCK OF CODE WHICH RUNS ONLY WHEN IT IS CALLED.
- 5. FETCHALL ():
- THIS FUNCTION WILL RETURN ALL THE ROWS FROM THE RESULT SET IN THE FORM OF A TUPLE CONTAINING THE RECORDS.
- 6. COMMIT():
- THIS FUNCTION PROVIDES CHANGES IN THE DATABASE PHYSICALLY.

MYSQL CODES:

```
mysql> USE library;
Database changed
mysql> show tables;
Empty set (0.26 sec)

mysql> use library;
Database changed
mysql> create table books (bookname varchar(50),authorname varchar(50),bookcode varchar(10),total int);
Query OK, 0 rows affected (25.77 sec)

mysql> show tables;
+-----+
| Tables_in_library |
+-----+
| books              |
+-----+
1 row in set (0.23 sec)

mysql> USE library;
Database changed
mysql> show tables;
+-----+
| Tables_in_library |
+-----+
| books              |
+-----+
1 row in set (0.74 sec)

mysql> create table issue(bookname varchar(50),bookcode int,studentname varchar(50), issuedate varchar(50));
Query OK, 0 rows affected (2.27 sec)
```

```
mysql> use library;
Database changed
mysql> CREATE TABLE `return` (
  ->     bookname VARCHAR(50),
  ->     bookcode INT,
  ->     studentname VARCHAR(50)
  ->     -- Additional columns and constraints can be added here
  -> );
Query OK, 0 rows affected (0.30 sec)
```

```
mysql> show tables;
+-----+
| Tables_in_library |
+-----+
| books              |
| issue              |
| return             |
+-----+
3 rows in set (0.00 sec)
```

PYTHON CODE :

```
import mysql.connector
con=mysql.connector.connect(host = 'localhost',

                             user ='root',
                             password='HIMANIDHAWAN10846',
                             database='library')
mycursor=con.cursor()
def donatebook():
    bookname=input('enter name of book you want to donate:')
    authorname=input('enter name of author:')
    bookcode=input('enter code of book:')
    total=int(input('how many copies of the book you want to donate:'))
    print('what to make more donation ?')
    a=input('enter your choice y for yes and n for no :')
    if a=='y' :
        donatebook()
    if a=='n' :
        mycursor=con.cursor()
        sql="INSERT INTO books (bookname,authorname,bookcode,total) VALUES(%s,%s,%s,%s)"
        val=(bookname,authorname,bookcode,total)
        mycursor.execute(sql,val)
        con.commit()
        print('THANKS FOR YOUR SUPPORT|')
```

```
def deletebook():
    bookcode=int(input('enter book code:'))
    a='delete from books where bookcode =%s;'
    data=(bookcode,)
    c=con.cursor()
    c.execute(a,data)
    con.commit()
    print('book deleted successfully')
def issuebook():
    bookname=input('enter name of the book:')
    studentname=input('enter name of the student :')
    bookcode=int(input('enter book code:'))
    issuedate=input('enter date:')
    print("WOULD YOU LIKE TO ISSUE ANOTHER BOOK ?")
    a=input('Enter y for yes and n for no:')
    if a=='y':
        issuebook()
    elif a=='n':
        mycursor=con.cursor()
        sql='insert into issue (bookname,bookcode,studentname,issuedate)values(%s,%s,%s,%s)'
        val=(bookname,bookcode,studentname,issuedate)
        mycursor.execute(sql,val)
        con.commit()
        print('book issued')
    else:
```



```
else:
    print('invalid response')
def issuedbooks():
    mycursor=con.cursor()
    mycursor.execute('select * from issue')
    myrecords=mycursor.fetchall()
    no_rec=mycursor.rowcount
    print('total no. of entries found are:',no_rec)
    for x in myrecords:
        print(x)
def returnbook():
    bookname=input('enter name of the book:')
    studentname=input('enter name of the student :')
    bookcode=int(input('enter book code:'))
    returndate=input('enter date:')
    print("WOULD YOU LIKE TO RETURN ANOTHER BOOK ?")
    a=input('Enter y for yes and n for no:')
    if a=='y':
        returnbook()
    elif a=='n':
        mycursor=con.cursor()
        sql='insert into `return` (bookname,bookcode,studentname,returndate)values(%s,%s,%s,%s)'
        val=(bookname,bookcode,studentname,returndate)
        mycursor.execute(sql,val)
        con.commit()
```

```
File Edit Format Run Options Window Help
con.commit()
print('BOOK RETURNED THANKYOU ')
else:
    print('INVALID RESPONSE')
def returnedbooks():
    mycursor=con.cursor()
    mycursor.execute('select * from `return`')
    myrecords=mycursor.fetchall()
    no_rec=mycursor.rowcount
    print('total no. of entries found are:',no_rec)
    for x in myrecords:
        print(x)
def displaybooks():
    mycursor=con.cursor()
    mycursor.execute('select * from books')
    myrecords=mycursor.fetchall()
    no_rec=mycursor.rowcount
    print('total no. of books found are:',no_rec)
    for x in myrecords:
        print(x)
def exit():
    for i in range (1):
        break
def choice():
    user_choice=input('enter your choice (1-8):')
```

```
def choice():
    user_choice=input('enter your choice (1-8):')
    if (user_choice == '1'):
        donatebook()
    elif (user_choice=='2'):
        deletebook()
    elif(user_choice=='3'):
        issuebook()
    elif(user_choice == '4'):
        issuedbooks()
    elif (user_choice=='5'):
        returnbook()
    elif (user_choice == '6'):
        returnedbooks()
    elif (user_choice=='7'):
        displaybooks()
    elif (user_choice == '8'):
        exit()
    else:
        print('invalid choice')
        choice()

def intro():
    print('HELLO USER welcome to our family')
    print('giving is not just about making a donation ,it is must making difference')
```

```
def intro():  
    print('HELLO USER welcome to our family')  
    print('giving is not just about making a donation ,it is must making difference')  
    print('whatever the cost of our libraries the price is cheap compared to that of an ignorant nation')  
    print('choose your choice wisely:')  
    print("""
```

LIBRARY MANAGER

```
1.DONATE BOOK  
2.DELETE BOOK  
3.ISSUE BOOK  
4.SHOW ISSUED BOOKS  
5.RETURN BOOK  
6.RETURNED BOOKS  
7.DISPLAY BOOKS  
8. EXIT PROGRAM""")  
    choice()
```

```
def password():  
    password=input('enter password:')  
    if password=='nvps':  
        intro()  
    else :  
        password()  
password()
```

RESULT

enter password: nvps
HELLO USER welcome to our family
giving is not just about making a donation ,it is must making difference
whatever the cost of our libraries the price is cheap compared to that of an ignorant nation
choose your choice wisely:

LIBRARY MANAGER

- 1.DONATE BOOK
- 2.DELETE BOOK
- 3.ISSUE BOOK
- 4.SHOW ISSUED BOOKS
- 5.RETURN BOOK
- 6.RETURNED BOOKS
- 7.DISPLAY BOOKS
8. EXIT PROGRAM

enter your choice (1-8):1

enter name of book you want to donate:the Queen of Jhansi

enter name of author:Mahasweta Devi

enter code of book:9788171195398

how many copies of the book you want to donate:10

what to make more donation ?

enter your choice y for yes and n for no :n

THANKS FOR YOUR SUPPORT

FOR DONATE ,
PASSWORD ,INTRO
FUNCTION

- EFFECT ON MYSQL:

```
mysql> use library;  
Database changed  
mysql> select * from books;
```

bookname	authorname	bookcode	total
wings of fire	APJ Abudul kalam Azad	9788173711466	5
the man who knew infinity	robert kanigel	9788817871693	100
the Queen of Jhansi	Mahasweta Devi	9788171195398	10

3 rows in set (0.59 sec)

- DELETE BOOK FUNCTION:

enter your choice (1-8):2
enter book code: 9788817871693
book deleted successfully

- EFFECT ON MYSQL:

```
mysql> select * from books;
```

bookname	authorname	bookcode	total
wings of fire	APJ Abudul kalam Azad	9788173711466	5
the Queen of Jhansi	Mahasweta Devi	9788171195398	10

2 rows in set (0.00 sec)

- **FOR ISSUING BOOKS:**

```
enter your choice (1-8):3
enter name of the book: APJ Abudul kalam Azad
enter name of the student :shivani
enter book code:9788173711466
enter date:30.06.2023
WOULD YOU LIKE TO ISSUE ANOTHER BOOK ?
Enter y for yes and n for no:n
book issued
```

- **EFFECT OF MYSQL:**

```
mysql> select * from issue;
```

bookname	bookcode	studentname	issuedate
the man who knew infinity	9788817871693	himani	2030-06-20
wings of fire	9788173711466	himani	2030-06-20
APJ Abudul kalam Azad	9788173711466	shivani	2030-06-20

```
3 rows in set (1.09 sec)
```

TO SEE ISSUED BOOKS:

enter your choice (1-8):4

total no. of entries found are: 3

('the man who knew infinity', '9788817871693', 'himani', datetime.date(2030, 6, 20))

('wings of fire', '9788173711466', 'himani', datetime.date(2030, 6, 20))

(' APJ Abudul kalam Azad', '9788173711466', 'shivani', datetime.date(2030, 6, 20))

• TO RETURN BOOK:

enter your choice (1-8):5

enter name of the book:wings of fire

enter name of the student :kaniska

enter book code: 9788173711466

enter date:01.07.2023

WOULD YOU LIKE TO RETURN ANOTHER BOOK ?

Enter y for yes and n for no:n

BOOK RETURNED THANKYOU

• EFFECT ON MYSQL:

```
mysql> select * from `return`;
```

bookname	bookcode	studentname	returndate
wings of fire	9788173711466	himani	2004-07-20
wings of fire	9788173711466	kaniska	2001-07-20

2 rows in set (0.04 sec)

- TO SEE RETURNED BOOKS DATA:

```
enter your choice (1-8):6
total no. of entries found are: 2
('wings of fire', '9788173711466', 'himani', datetime.date(2004, 7, 20))
('wings of fire', '9788173711466', 'kaniska', datetime.date(2001, 7, 20))
```

- TO DISPLAY BOOKS:

```
enter your choice (1-8):7
total no. of books found are: 2
('wings of fire', 'APJ Abudul kalam Azad', '9788173711466', 5)
('the Queen of Jhansi', 'Mahasweta Devi', '9788171195398', 10)
```

- TO EXIT PROGRAM;

LIBRARY MANAGER

```
1.DONATE BOOK
2.DELETE BOOK
3.ISSUE BOOK
4.SHOW ISSUED BOOKS
5.RETURN BOOK
6.RETURNED BOOKS
7.DISPLAY BOOKS
8. EXIT PROGRAM
enter your choice (1-8):8
```